

Black spruce reforestation in *Kalmia* heath: seedling response to forest floor mixing and mycorrhizal inoculation with *Paxillus involutus*

Gregg R. Walker and Azim U. Mallik

Abstract: In the boreal forests of eastern Canada, black spruce (*Picea mariana* (Mill.) Britton, Sterns, Poggenb.) regeneration often fails in nutrient-poor sites dominated by *Kalmia angustifolia* L. occurring after nonsevere forest fire, clearcut harvesting, and insect defoliation. Traditional silvicultural treatments have limited success in regenerating these sites. We tested whether forest floor mixing and planting black spruce seedlings preinoculated with an ectomycorrhizal (ECM) fungus, *Paxillus involutus* (Batsch) Fr., has potential to improve black spruce growth in *Kalmia*-dominated sites. We measured growth and foliar nutrient concentrations of ECM-inoculated and control black spruce seedlings planted in burned and insect-defoliated sites after forest floor mixing in Terra Nova National Park, Newfoundland. We also measured soil moisture, temperature, pH, total phenol and nutrient concentrations, litter decomposition, and soil respiration in those sites. Seedlings in soil-mixed plots had significantly higher growth and foliar nutrient concentrations than nonmixed plots, likely because soil mixing improved the substrate conditions by increased soil moisture, pH, and litter decomposition. Seedlings in insect-defoliated sites had significantly higher increases in height and foliar nutrient concentrations than those in the burned sites. Inoculated seedlings had significantly higher initial ECM formation than the control seedlings, but they failed to achieve higher growth.

Résumé : Dans les forêts boréales de l'est du Canada, la régénération d'épinette noire (*Picea mariana* (Mill.) Britton, Sterns, Poggenb.) est souvent insuffisante sur les stations pauvres en nutriments dominées par *Kalmia angustifolia* L. qui s'établit à la suite d'un feu de faible intensité, d'une coupe à blanc ou d'une défoliation par les insectes. Les traitements sylvicoles traditionnels sont peu efficaces pour régénérer ces stations. Nous avons assumé que le scarifiage et la plantation de semis d'épinette noire pré-inoculés avec un champignon ectomycorhizien (ECM), *Paxillus involutus* (Batsch) Fr., pouvaient améliorer la croissance de l'épinette noire sur les stations dominées par le *Kalmia*. Nous avons mesuré la croissance et la concentration foliaire en nutriments de semis d'épinette noire inoculés et non inoculés avec le champignon ECM, puis plantés sur des sols scarifiés dans des stations qui avaient été brûlées ou défoliées par les insectes dans le parc national Terra-Nova, à Terre-Neuve. Nous avons aussi mesuré l'humidité du sol, la température, le pH, les concentrations en phénol et en nutriments, la décomposition de la litière et la respiration du sol de ces stations. La croissance des semis et la concentration en nutriments étaient significativement plus élevées dans les placettes scarifiées que dans les placettes non scarifiées, probablement parce que le scarifiage a amélioré l'état du substrat en augmentant l'humidité du sol, le pH et la décomposition de la litière. La croissance en hauteur et la concentration foliaire en nutriments des semis plantés sur des stations défoliées par les insectes étaient significativement plus élevées que celles des semis plantés sur des stations brûlées. La formation initiale d'ECM était significativement plus importante sur les semis inoculés que sur les semis témoins, mais leur croissance n'était pas meilleure.

[Traduit par la Rédaction]

Introduction

Forests dominated by black spruce (*Picea mariana* (Mill.) Britton, Sterns, Poggenb.) on medium- to poor-quality sites in eastern Canada particularly in central Newfoundland, northern Quebec, and Labrador commonly contain *Kalmia*

angustifolia L. (hereafter *Kalmia*) in the understory (Meades and Moores 1989). After forest canopy removal by low-severity fires, clearcut harvesting, or insect defoliation, black spruce regeneration can be limited in the presence of persistent heaths dominated by *Kalmia* (Meades 1986; Mallik 1995). This conversion of forest to *Kalmia*-dominated heath after natural or anthropogenic disturbance is a serious problem for the forest industry because it represents a loss of productive land base (Meades 1986). Park and protected area managers are also concerned that changing disturbance regimes are causing overrepresentation of heath to the detriment of landscape-scale biodiversity (Power 1996). Practical methods must be found to reverse this process to restore forest productivity and ecological integrity of the system.

Silvicultural techniques, such as brush cutting and prescribed burning, are not effective in improving black spruce

Received 23 April 2008. Resubmitted 15 January 2009.
Accepted 10 July 2009. Published on the NRC Research Press Web site at cjfr.nrc.ca on 27 October 2009.

G.R. Walker and A.U. Mallik,^{1,2} Department of Biology, Lakehead University, 955 Oliver Road, Thunder Bay, ON P7B 5E1, Canada.

¹Corresponding author (e-mail: azim.mallik@gmail.com).

²Present address: Mount Revelstoke and Glacier National Parks, Parks Canada Agency, Box 350, Revelstoke, BC V0E 2S0, Canada.

growth in *Kalmia*-dominated sites. Scarification studies conducted elsewhere have produced some encouraging results, but scalping was found to be ineffective (Prevost 1994; Thiffault et al. 2004). Scarification and spot fertilization at planting was found to improve black spruce growth in *Kalmia*–*Vaccinium*-dominated sites (Thiffault et al. 2004, 2005, 2005). Thiffault and Jobidon (2006) expressed optimism for using these methods for reforesting ericaceous heath. However, others suspect that these methods may not fully release black spruce seedlings in the long run from dense *Kalmia* because the inhibitory effects of *Kalmia* humus may persist (Bradley et al. 1997; Wallstedt et al. 2002). Likewise, fertilization may temporarily improve nutrient availability for black spruce, but the potential directly inhibitory effects of *Kalmia* leachates on black spruce and its ectomycorrhizae may persist for an extended period (Yamasaki et al. 2002). LeBel et al. (2008) reported that the removal of *Kalmia* increases black spruce seedling growth via improved forest floor nutrients, but the positive effect of fertilizer remains short lived. Joannis et al. (2007) reported that soil enzyme inhibition by the condensed tannins of *Kalmia* may play an important role in the mechanism of black spruce growth inhibition in *Kalmia* heath.

Considering the characteristics of *Kalmia* sites and autecology of black spruce, we thought of using two methods for maximizing black spruce growth in *Kalmia*-dominated heath. First, mechanical mixing of *Kalmia* humus, which would control *Kalmia* by destroying its rhizomatous buds (Mallik 1991; Keenan et al. 1994; Frey et al. 2003) and, concomitantly, improve soil conditions by increasing decomposition and nutrient availability (Messier et al. 1995; Prescott et al. 2000). After mulching, increased microbial activity and decreased inputs of *Kalmia* leaf litter might also reduce *Kalmia* allelochemicals, such as phenolics and condensed tannins. Several authors have reported enhanced conifer growth after forest floor mixing (Burger and Pritchett 1988; Messier et al. 1995), but this has not been tested in *Kalmia*-dominated sites.

As an adjunct to the forest floor mixing approach, we used a second method by planting black spruce seedlings preinoculated with an ectomycorrhizal (ECM) fungus in *Kalmia* sites. Several authors have shown that planting tree seedlings preinoculated with ECM fungi can be an effective means of forest regeneration in nutrient-poor sites (Marx and Cordell 1988; Villeneuve et al. 1991). The inoculated seedlings may have better survival and growth because of increased nutrient uptake, tolerance to water stress, and resistance to fungal pathogens (Perry and Choquette 1987). Zeng and Mallik (2006) reported that, under controlled conditions, the ECM fungus *Paxillus involutus* (Batsch) Fr. can degrade *Kalmia* phenolics and use the degradation products as their carbon source. Thus, black spruce inoculated with ECM fungi in *Kalmia* sites may have improved nutrient availability and reduced allelopathic effects of *Kalmia*. Mallik et al. (1998) reported that *P. involutus* (isolate NF4) forms abundant ECM with black spruce in *Kalmia* soils under greenhouse conditions. In that study, the inoculated black spruce exhibited enhanced growth in soils amended with *Kalmia* leachates and in the presence of living *Kalmia*. However, uncertainties remain in that seedlings inoculated with ECM fungi may not always experience enhanced

growth because of multiple site factors (Bledsoe et al. 1982; Molina and Chamard 1983). Although black spruce inoculated with ECM fungi seem to overcome *Kalmia*-induced growth inhibition under controlled conditions, growth effects have not been tested under field conditions.

Our objectives for this study were (i) to determine the effect of forest floor mixing and ECM inoculation on growth and foliar nutrient concentrations of black spruce in *Kalmia*-dominated sites and (ii) to determine changes in soil physical and chemical properties due to the treatments. We hypothesized that increased soil temperature, litter decomposition, pH, and nutrient availability through forest floor mixing and increased nutrient uptake through *P. involutus* inoculation would result in higher seedling growth and foliar nutrient concentrations of planted black spruce.

Materials and methods

Study area

We conducted this study in Terra Nova National Park (TNNP) (48.5°N, 54.0°W, elevation 100 m a.s.l.), approximately 90 km southeast of Gander, Newfoundland. This area belongs to the eastern boreal forest region (Rowe 1972). The study sites were dominated by black spruce and balsam fir (*Abies balsamea* (L.) Mill.) with some tamarack (*Larix laricina* (Du Roi) K. Koch), white birch (*Betula papyrifera* Marsh.), trembling aspen (*Populus tremuloides* Michx.), and red maple (*Acer rubrum* L.). The forest understory consisted primarily of ericaceous shrubs, dominated by *Kalmia* (38%–81% cover) and *Vaccinium angustifolium* Ait., *Ledum groenlandicum* Oeder, with *Rhododendron canadense* (L.) Torr. as subdominants. Other common ground layer plants include *Cornus canadensis* L., *Prunus pensylvanica* L.f., *Pteridium aquilinum* (L.) Kuhn, mosses (*Pleurozium schreberi* (Brid.) Mitt., *Ptilium crista-castrensis* (Hedw.) De Not.), and lichens (*Cladonia* and *Cladina* spp.). The forest soils in the area are Humo-Ferric Podzols (Soil Classification Working Group 1998) formed on a moderately well-drained fluvio-glacial deposit with a 5–10 cm humus layer.

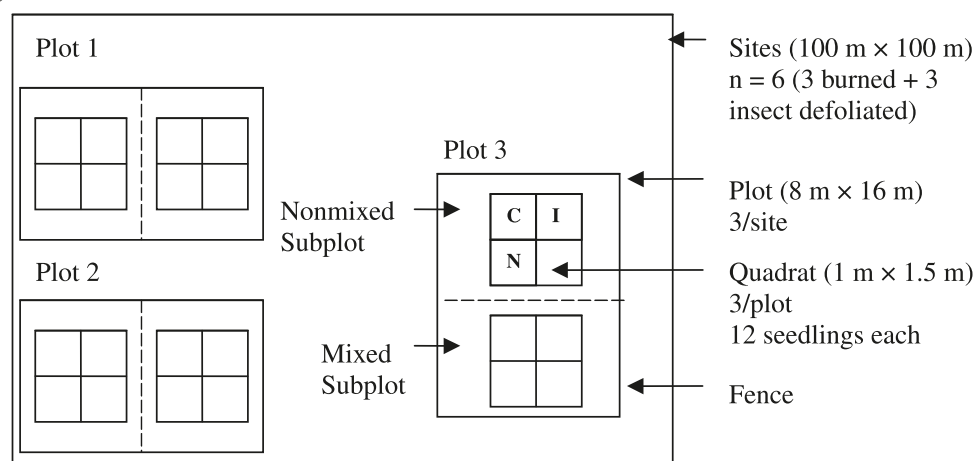
Study sites

We studied *Kalmia*-dominated sites originating from two types of common natural disturbances in the area: forest fire and insect defoliation by spruce budworm (*Choristoneura fumiferana* Clem.). Although characteristics of each fire would have been different, they all produced two common effects: (i) killed all trees and understory plants creating an open canopy condition and (ii) postfire *Kalmia*-dominated vegetation. On the other hand, the insect-defoliated sites retained partial canopy cover because of standing dead and partially damaged canopy trees with intact understory. *Kalmia* become dominant after both types of disturbance because of its efficient vegetative regeneration from belowground rhizomes and stem bases. Three experimental sites were established in each disturbance type. The insect-defoliated sites (Bluehill 1, 2, and 3) were infested by spruce budworm between 1977 and 1983 (20–26 years ago). Because of tree mortality, these sites had canopy openings ranging from 0.25 to 0.5 ha. The wildfire sites were selected to represent the variability associated with a range of times

Table 1. Characteristics of trees and regeneration in three burned and three insect-defoliated experimental sites.

Parameter	Insect defoliated			Burned		
	Bluehill 1	Bluehill 2	Bluehill 3	Terra Nova	Ochre Hill	Dunphy's Pond
Stem density (no.·100 m ⁻²)	8.8 (3.8)	18.4 (4.4)	19.2 (2.8)	2.8 (1.3)	33.6 (4.3)	35.6 (12.3)
Basal diameter (cm)	13.2 (4.7)	6.6 (2.2)	4.4 (1.8)	2.3 (0.9)	2.3 (0.1)	3.0 (0.7)
Canopy closure	12.3 (4.9)	3.5 (2.2)	11.3 (3.4)	0 (0)	0 (0)	5 (5)
Species composition (% total basal area)						
Black spruce	47.9 (21.6)	59.8 (22.7)	22.5 (19.5)	80.0 (20.0)	75.6 (14.0)	100.0 (0.0)
Balsam fir	31.0 (19.0)	1.9 (0.8)	35.7 (21.7)	—	0.3 (0.3)	—
Tamarack	—	—	—	—	24.1 (14.1)	—
Red maple	4.1 (4.1)	19.3 (19.3)	—	—	—	—
Trembling aspen	—	—	41.8 (23.4)	—	—	—
White birch	16.9 (16.9)	19.0 (19.0)	—	—	—	—
Black spruce seedling density (no.·100 m ⁻²)	2.0 (1.6)	8.0 (2.5)	3.2 (1.6)	1.6 (0.7)	24.0 (2.2)	16.8 (12.3)

Note: Values are means, with SEs given in parentheses, of circular quadrats (50 m²) ($n = 5$). Black spruce seedlings are ≤ 3 cm basal diameter. —, Species absent.

Fig. 1. Layout of experimental design in the burned and insect-defoliated sites. Quadrats were planted with control (C), inoculated (I), and nursery (N) seedlings.

since disturbance by fire. Wildfires were suppressed at the three burned sites (Terra Nova, Ochre Hill, and Dunphy's Pond). The fire at Terra Nova (314 ha) occurred 24 years ago, the small Ochre Hill fire (2 ha) occurred 33 years ago, and the Dunphy's Pond fire (800 ha) occurred 42 years ago (Power 1996). We determined canopy tree and seedling density, tree species composition, and basal area in each of the burned and insect-defoliated sites from five 50 m² circular plots prior to the experiment. The insect-defoliated sites had both remnant canopy trees and regenerating saplings and higher tree species richness with more deciduous species than the burned sites, whereas the burned sites contained only infrequent tree seedlings and saplings (Table 1). Canopy closure was significantly greater in the insect-defoliated sites than in the burned sites ($t = 2.6$; $p < 0.01$). In all sites, the density of black spruce seedlings was low (mean 160–2400 stems·ha⁻¹) relative to the standard 3000 stems·ha⁻¹ considered necessary for black spruce forest regeneration (Newton and Weetman 1993). Forest floor humus thickness at the study sites ranged from 7.6 to 16.4 cm. All the insect-defoliated and burned sites were dominated by *Kalmia* and

had been for at least 20 years following disturbance. Before disturbance, all the study sites consisted of closed canopy black spruce forests, which was evident from the forest resource inventory data.

Experimental design

We established three randomly selected sites (whole blocks) nested in each of the two disturbance types: insect defoliated and burned. All study sites were located within 3 km of access roads, and distances between sites ranged from 0.8 to 20 km. Within each site, three 4.5 m × 8 m replicate plots (subblocks) were randomly located (Fig. 1). The plots were located in the open heath at least 10 m from the forest edge. We erected a wire-mesh fence around each plot to protect the seedlings from small mammalian herbivores. Each plot was divided into two equal halves (subplots), and the forest floor humus in one-half was mixed (described below). Each subplot was further divided into four 1 m × 1 m quadrats. Three of these quadrats were planted with 12 black spruce seedlings from each of the seedling groups with 0.25 m spacing between them. The quadrats were separated

from one another and from the surrounding vegetation by a 1 m buffer. Seedlings in the adjacent mixed and nonmixed subplots were separated by 2 m.

Forest floor mixing

Forest floor mixing was performed during 27–30 June 2002 with a rotary-style mixing machine (Craftsman Rototiller, 5.5 horsepower) specially modified to increase its efficiency in stony soils and tough intertwining *Kalmia* rhizomes. The aboveground vegetation was first cut with a power brush saw, then the forest floor (L, F, and H horizons) and belowground plant parts were mixed to the depth of the Ae mineral soil.

Seedling culture and treatments

To evaluate the response of black spruce seedlings to ECM inoculation, we grew the seedlings using low levels of fertilizer because high soil nutrients may discourage ECM formation (Danielson et al. 1984). These seedlings were small and nutritionally stressed when outplanted. To assess whether the responses of the seedlings to soil mixing were an artifact of the small, nutrient-deficient inoculated seedlings, we included another batch of seedlings grown at a commercial nursery that were exposed to high levels of fertilization to promote growth after outplanting (Malik and Timmer 1996).

The first two groups of seedlings (inoculated and control) were cultured for 32 weeks (26 weeks growth and 6 weeks cold hardening) at Lakehead University greenhouse, Thunder Bay, Ontario. Black spruce seeds were obtained from Wooddale Provincial Tree Nursery (Grand Falls, Newfoundland), surface sterilized with 2% NaClO₃, germinated in Petri dishes, and sown in 112-well multipots with (well volume 128 mL). The growth medium was a 1:1 mixture of peat and vermiculite, which was autoclaved to reduce contamination. After emergence, the seedlings received 20:8:20 (N:P:K) fertilizer (Plant Products, Brampton, Ontario) in the irrigation water on alternate days at an N concentration of 100 ppm. Five weeks later, these seedlings were split into two groups. One group of approximately 500 randomly selected seedlings was inoculated with *P. involutus* (isolate 211804) obtained from the Canadian Collection of Fungal Culture, Ottawa, Ontario. In a greenhouse experiment, this isolate was to be well adapted to *Kalmia* soil (Zeng and Mallik 2006). Germinated black spruce seedlings were transplanted into an inoculated peat and vermiculite growth medium, prepared by combining two methods of inoculation to enhance the likelihood of success. First, *P. involutus* was cultured in 25 Petri dishes in the laboratory on sterile modified Melin–Norkans growth medium (Danielson 1982). Surface-sterilized jack pine (*Pinus banksiana* Lamb.) seeds (four or five) were added to improve the growth rate of *P. involutus* (Zeng et al. 2003). These cultures were then incubated at 25 °C for 2 weeks. Using sterile technique, *P. involutus* mycelia were carefully separated from the growth medium and then mixed in a blender (Waring Corp., Hartford, Connecticut) with 750 mL of sterile water to produce uniform inoculation solution. Second, a vermiculite inoculum was produced by adding 1 L of modified Melin–Norkans growth medium to 10 L of vermiculite in previously autoclaved 1 L glass jars and incubating at 25 °C for

8 weeks. The growth medium was removed from the vermiculite inoculum by wrapping the mixture in four-ply cheesecloth (Veratec International Paper, Stamford, Connecticut) and rinsing with water for 5 min. One litre sterile glass jars were filled with the washed vermiculite inoculum, 10 mL of the blended liquid inoculum was added to each jar, and the jars were incubated for 5 days at 25 °C. An extensive network of the ECM mycelia was visible in the vermiculite–water inoculum. The resultant inoculum was mixed into an autoclaved seedling growth medium of mixed peat and vermiculite (1:1). Approximately 500 randomly selected seedlings were transplanted into this inoculated growth medium. The second group of approximately 500 seedlings (control) was transplanted into sterile growth medium without ECM. After transplanting, both the inoculated and control seedlings received 20:20:20 (N:P:K) fertilizer at an N concentration of 100 ppm twice monthly (Walker 2005). Environmental conditions during the growth and cold hardening periods followed Landis et al. (1990).

The third group of seedlings (commercial nursery) was grown for 22 weeks (14 weeks growth and 8 weeks cold hardening) at Wooddale Provincial Tree Nursery under conditions similar to those of the inoculated and control seedlings. However, these seedlings received NPK fertilizer in the irrigation water (alternate days) at an N concentration varying from 35 to 150 ppm throughout the growing period. To describe the morphology of the seedlings prior to outplanting, we measured shoot height, root-collar diameter, oven-dry mass (root, shoot, and shoot/root ratio), and ECM formation (described below) of eight randomly selected seedlings.

Shoot height, root-collar diameter, and dry mass

For shoot height and root-collar diameter, six randomly selected seedlings were measured from each quadrat immediately after outplanting (1–10 July 2002), and they were re-measured 14 months later at the end of the second growing season (early September 2003) (648 seedlings). To determine oven-dry mass, three randomly selected seedlings were obtained per quadrat before outplanting and at the end of the second growing season (324 seedlings). These seedlings were carefully excavated from soil, placed in plastic bags and kept cool (4 °C) during transport. Shoots (stem and needles) were separated from the roots, air-dried, and stored in paper bags at room temperature (22 °C). The root systems were gently washed with tap water to remove soil and stored in a refrigerator (4 °C) for determining ECM formation. Subsequently, they were air-dried (22 °C). Roots and shoots were oven-dried at 70 °C for 3 h (Allen 1989) and weighed with an electronic balance. Change in shoot/root ratio was calculated as the difference between final and initial shoot/root ratio.

Foliar nutrient concentration

We collected three randomly selected seedlings from each quadrat after the second growing season (324 seedlings), placed them in plastic bags and kept them cool (4 °C) during transport, and then air-dried and stored in paper bags at room temperature (22 °C). Foliage from the three subsamples from each quadrat was ground in a mill and mixed together to form one composite sample. We measured total

nitrogen (N) as absorbance (630 nm) using a spectrophotometer (Skalar auto analyzer, Skalar Inc., Norcross, Georgia) after digesting 0.2 g samples by the Kjeldahl method. Total phosphorus (P) and potassium (K) were determined by inductively coupled argon plasma spectrometry (Varian Vista Pro ICAP Radial; Varian Inc., Mississauga, Ontario) after acid digestion (HNO_3 , HCl , H_2O_2) of 0.5 g samples (Kalra and Maynard 1991).

Ectomycorrhizal formation

We used the root systems of three randomly selected seedlings per quadrat at the end of the first and second growing seasons (324 seedlings). After soil had been removed by washing, root systems were cut into 2 cm segments to allow separation of roots without damaging ECM. Counts of ECM and non-ECM short roots were made on 10 randomly selected root segments from each root system at $10\text{--}45\times$ magnification using a dissecting stereomicroscope. From these counts, we determined percentages of total short roots that had ECM. We also used this method to evaluate ECM formation on each of the seedling groups immediately before outplanting. We morphotyped the ECM-inoculated seedlings prior to outplanting by comparing these with published ectomycorrhizal morphotype descriptions of *P. involutus* and confirmed that the ECM on the preoutplanted seedlings were *P. involutus* (Ingleby et al. 1990). We deemed morphotyping of ECM to be an adequate method of identifying *P. involutus* because a recent comparative study of morphotyping and molecular biological techniques showed that these methods are complementary and give similar results for most common ECM genera (Sakakibara et al. 2002). At the end of the experiment, we quantified the level of total mycorrhizal and nonmycorrhizal short roots in the outplanted seedlings to relate their overall mycorrhization with growth and foliar nutrient contents. Because our ECM inoculated seedlings produced no significant gain in growth compared with the other treatments, we did not morphotype the ECM inoculated seedlings.

Soil physical properties

We measured volumetric soil moisture and soil temperature at five random locations in each subplot during the second growing season (23 August 2003) (180 subsamples, $n = 36$). Measurements were made at 6–8 cm depth within an 8–10 h period, beginning 48 h after a sustained rainfall. We used an electronic probe and meter (Thetaprobe ML2x and Thetameter HH2; Delta-T Devices Ltd., Cambridge, UK) for soil moisture and a thermocouple and meter (Tri-sense KX-37000-00; Cole-Parmer Instrument Co., Vernon Hills, Illinois) for soil temperature.

Litter decomposition

We installed three replicate litterbags in each subplot at the beginning of the first growing season to determine decomposition rate (108 litterbags). Litter consisting of leaves, twigs, and mosses from a *Kalmia*-dominated site was collected and air-dried for 1 week. Litter bags were constructed using two sheets (20 cm $\times$ 20 cm) of 1 mm fiber glass mesh fastened together using inert alloy staples. We then filled the bags with 10 g air-dry litter. At this time, 10 g samples of air-dry litter were retained to estimate the initial oven-dry

mass of the litterbags. Litterbags were placed on the fermentation layer of the forest floor, and inert alloy pins were used to fix them in position. Some litterbags were transported to the field sites but were not installed to estimate mass loss during transportation. Litterbags were collected at the end of the second growing season (410 days after installation). These were gently cleaned of external debris, placed in plastic bags and kept cool (4 °C) during transport, air-dried, and stored in paper bags at room temperature (22 °C), oven-dried at 105 °C to constant mass (5 h), and weighed on an electronic balance. We measured soil respiration at five random locations in each subplot at the end of the second growing season (180 samples). We used an infrared gas analyzer (Nortech, Ottawa, Ontario), which measured CO_2 evolution in a sampling chamber covering 78.5 cm² of soil surface. The dry litter layer was brushed aside, and the sampling chamber was inserted 1 cm deep into the forest floor.

Soil chemical properties

We sampled soil from four random locations in each subplot at the end of the second growing season (September 2003) (144 samples). We excavated separate samples of forest floor humus (L, F, and H layers) and mineral soils (Ae and B) from approximately 10 cm $\times$ 10 cm area; each sample was approximately 200 g. The soil samples were placed in plastic bags and kept cool (4 °C) during transport, air-dried, and stored in paper bags at room temperature (22 °C). Air-dry soil samples were sieved (2 mm), and equal volumes of the four samples from each subplot were mixed into one composite sample ($n = 36$). Air-dry moisture content was determined by weighing 1 g samples on an electronic balance before and after oven-drying them at 105 °C to constant mass (5 h). Air-dry soils were used for all analyses, and the results were converted to oven-dry mass.

All soil samples were analyzed for pH, total water-soluble phenols, exchangeable inorganic ions (Ca, K, Mg, Na, Al, Cu, Fe, Mn, and Zn), total nitrogen (total N), ammonium nitrogen ($\text{NH}_4\text{-N}$), and inorganic phosphorus ($\text{PO}_4\text{-P}$). Soil pH was measured with a potentiometer (Accumet pH probe; Fischer Scientific, Nepean, Ontario) after shaking air-dry soil in distilled water. For forest floor humus, the soil:distilled water ratio was 1:4; for mineral soils, it was 1:2 (Kalra and Maynard 1991). Total phenolic concentration of soils was determined by comparing extractions (distilled water) against a standard curve developed using phenol crystals (Swain and Hillis 1959). We used Folin–Ciocalteu reagent and measured absorbance (725 nm) of the filtrate using a spectrophotometer (Skalar auto analyzer; Skalar Inc.).

Soil cations were extracted with ammonium acetate (pH 7.0) and 0.1 mol·L⁻¹ HCl was used for metals (Allen 1989). After extraction, cations and metals were measured by inductively coupled argon plasma spectrometry (Varian Vista Pro ICAP Radial). We measured total N as absorbance (630 nm) using a spectrophotometer after digestion of 0.2 g samples by the Kjeldahl method. The $\text{NH}_4\text{-N}$ was extracted with 1.0 mol·L⁻¹ KCl and measured as absorbance (535 nm) using a spectrophotometer. The $\text{PO}_4\text{-P}$ was extracted using the Bray P1 method and measured as absorbance (882 nm) with a spectrophotometer (Kalra and Maynard 1991). For all extractions, we used 1.5 g soil samples and Whatman

Table 2. Mean preplanting shoot height, root-collar diameter, ectomycorrhizal (ECM) formation, oven-dry mass (shoot and root), and shoot/root ratio of three groups of black spruce seedlings ($n = 24$).

Seedling group	Shoot height (cm)	Root-collar diameter (mm)	ECM formation (short roots·10 cm ⁻¹)	Root dry mass (g)	Shoot dry mass (g)	Shoot/root ratio
Inoculated	10.1c	1.62b	47a	0.150b	0.336b	1.83b
Nursery	21.8a	2.15a	13b	0.399a	1.512a	3.05a
Control	15.1b	2.09a	2c	0.114b	0.432b	3.20a

Note: Within columns, values with different letters are significantly different at $p < 0.05$.

No. 5 filter paper. Soil samples were analyzed for total C using finely ground and sieved (1 mm) samples in a dry combustion carbon analyzer (LECO analyzer; Laboratory Equipment Corporation, St. Joseph, Michigan).

Data analysis

Differences in seedling characteristics immediately before outplanting were determined by analysis of variance (ANOVA) followed by Scheffé's multiple contrasts ($n = 24$). The exception was with preplanting ECM formation, which was analyzed using a Kruskal–Wallis test for non-parametric data (with unequal variances) followed by Dunnett's T3 multiple comparisons because these data were not normally distributed.

Increases in seedling shoot height, root-collar diameter, dry mass, ECM formation, and foliar nutrient concentration were analyzed with a three-factor ANOVA for nested designs followed by Scheffé's multiple contrasts ($n = 108$ experimental units, 324 subsampled seedlings (648 for shoot height and root-collar diameter)). Relative increases were used to account for differences between treatment groups in initial seedling size. We used a nonbiased equation for relative growth rates (Hoffmann and Poorter 2002):

$$\text{Relative height increase} = \text{mean}(\ln \text{height}_{\text{initial}}) - \text{mean}(\ln \text{height}_{\text{final}})$$

Relationships between foliar nitrogen and relative increase in shoot dry mass of seedlings were tested using linear regression ($n = 36$ experimental units, 108 subsampled seedlings).

Soil physical and chemical property and litter decomposition data were analyzed using a two-factor ANOVA for nested designs followed by Scheffé's multiple contrasts ($n = 36$ experimental units, 180 subsamples (108 for litterbags, 144 for soil chemical properties)).

Data distributions were examined for normality using the Shapiro–Wilk statistic, and variances were tested for homogeneity using Levene's statistic. Foliar N data were transformed using a reciprocal function. For soil variables that were not distributed normally or that had unequal variances, appropriate transformation (square root, log₁₀, reciprocal, or 1/square root) was applied prior to analysis. We presented back-transformed means and standard errors. Data analyses were performed using Datadesk (version 6.0; Data Description Inc., Ithaca, New York) and SPSS software (version 9.0; SPSS Inc., Chicago, Illinois).

Results

Seedling characteristics prior to outplanting

Prior to outplanting, shoots of the commercial nursery-

grown seedlings were 44% longer than control seedlings (Scheffé's test, $S = 7.7$, $p < 0.001$) and shoots of inoculated seedlings were 33% shorter than control seedlings ($S = 5.6$, $p < 0.001$; Table 2). However, root-collar diameters of nursery seedlings were not significantly different from those of control seedlings. Mean root-collar diameter of inoculated seedlings was 22% less than control seedlings ($S = 3.8$, $p < 0.001$). The nursery seedlings had more than twice as much shoot and root dry mass compared with the control ($S = 10.9$, $p < 0.001$ and $S = 6.9$, $p < 0.001$, respectively), but the inoculated seedlings were not significantly different from the control. The shoot/root dry mass ratio of nursery seedlings was not significantly different from control, but the ratios of the inoculated seedlings were 43% lower than the control seedlings ($S = 4.0$, $p < 0.01$). Nursery seedlings had significantly higher levels of ECM formation (13 short roots·10 cm⁻¹) than control seedlings (2 short roots·10 cm⁻¹) (Dunnett's T3 test, $T = 1.9$, $p < 0.001$), but the inoculated seedlings had the highest number of mycorrhizal short roots (47 short roots·10 cm⁻¹) ($T = 8.1$, $p < 0.001$). The ECM morphotype on the inoculated seedlings was *P. involutus*, whereas those on the control seedlings appeared to be other morphotypes.

Shoot height

Relative increases in shoot height of control and inoculated seedlings were significantly greater in the insect-defoliated sites than in the burned sites, but the disturbance type had no significant effect on relative increase in shoot height of nursery seedlings ($F = 7.5$, $p < 0.01$; Fig. 2a, Table 3). There was no significant difference in relative increase of shoot height after two growing seasons between the control and inoculated seedlings either in burned or insect-defoliated sites with or without soil mixing.

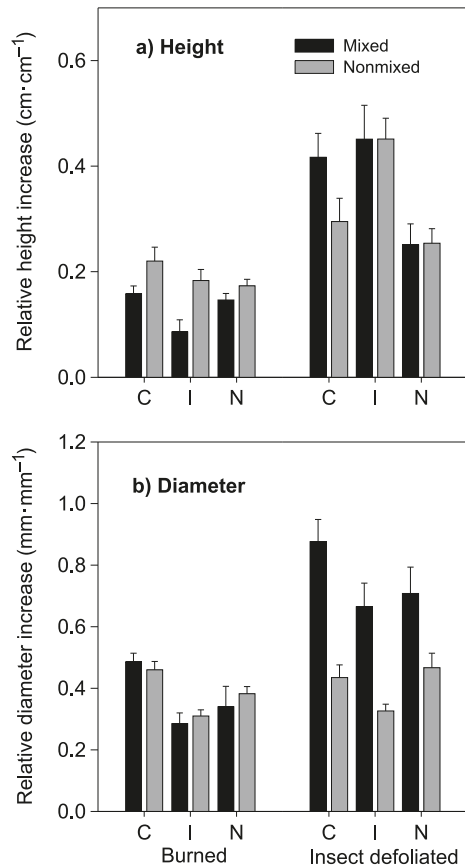
Root-collar diameter

In insect-defoliated sites, all seedling types had significantly greater relative increase in diameter in soil-mixed plots than in nonmixed plots ($F = 12.9$, $p < 0.02$) (Fig. 2b, Table 3). In burned sites, soil mixing had no significant effect on relative diameter increase for control, nursery, or inoculated seedlings compared with the nonmixed plots. For all seedling types, relative diameter increase was largest in mixed plots of insect-defoliated sites. Control seedlings in the second growing season had significantly greater relative increase in diameter than the inoculated seedlings in both burned and insect-defoliated sites ($S = 4.4$, $p < 0.01$, and $S = 4.0$, $p < 0.01$, respectively).

Shoot dry mass

Overall, the relative increase in shoot dry mass was

Fig. 2. Relative increase in shoot height (a) and root-collar diameter (b) of control (C), inoculated (I), and nursery (N) black spruce seedlings after two growing seasons. Bars are means, and error bars are SEs ($n = 108$). See Table 3 for ANOVA p values.



greater in insect-defoliated sites than in burned sites ($F = 8.5$, $p < 0.04$; Fig. 3a, Table 3). Control and inoculated seedlings in mixed plots of insect-defoliated sites had the largest relative increase in shoot mass (1.32 and 1.19 $\text{g}\cdot\text{g}^{-1}$, respectively). In insect-defoliated sites, relative increase in shoot dry mass was significantly higher in mixed plots than in nonmixed plots for all seedling types ($F = 7.2$, $p < 0.047$). In burned sites, forest floor mixing had no significant effect on relative increase in shoot dry mass for any of the seedling types. In the second growing season, control seedlings had significantly greater relative increase in shoot dry mass than the inoculated seedlings in burned sites ($S = 6.3$, $p < 0.001$), but there was no significant difference between these seedling types in the insect-defoliated sites.

Root dry mass

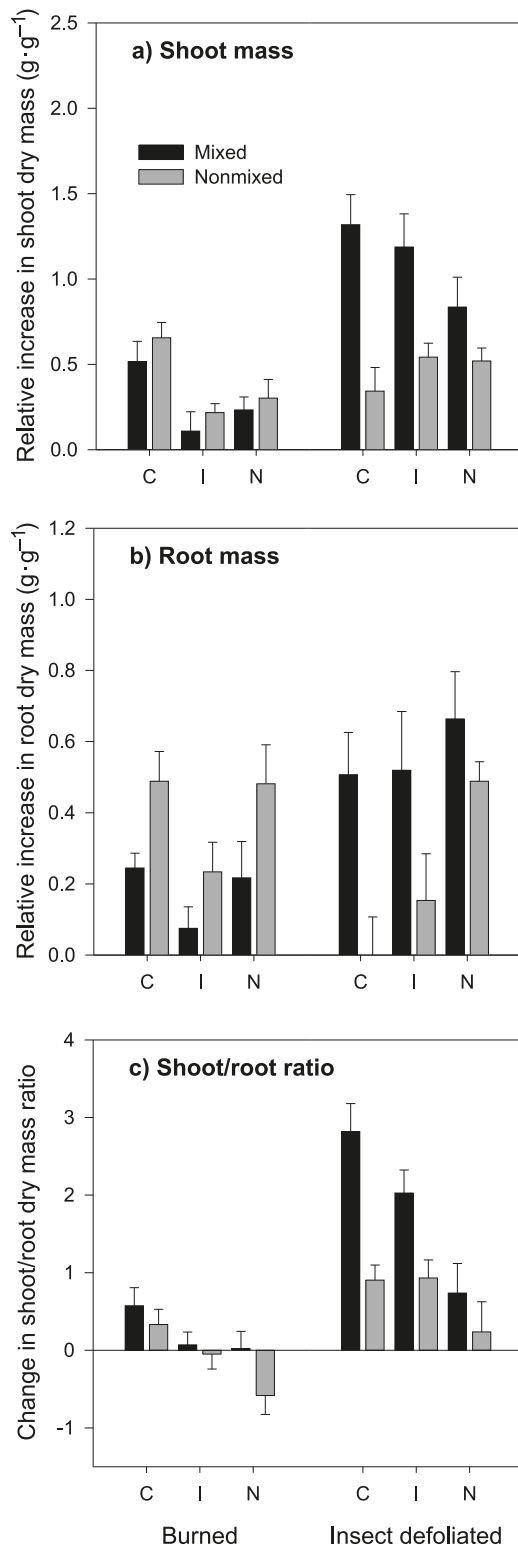
In insect-defoliated sites, the relative increase in root dry mass was significantly higher in mixed plots than nonmixed plots for all seedling types ($S = 2.8$, $p < 0.05$; Fig. 3b, Table 3). However, in burned sites, forest floor mixing did not result in significantly different relative increases in root dry mass for any seedling groups ($S = 1.8$, $p < 0.15$). In the second growing season, inoculated seedlings in burned sites had a significantly lower relative increase in root dry mass than the control seedlings ($S = 3.4$, $p < 0.03$), but there was no

Table 3. ANOVA results (p values) of seedling response variables.

Source of variation	df		Dry mass				Foliar nutrient concentration				ECM formation
	Numerator	Denominator	Shoot	Root	Shoot/root	N	P	K			
Disturbance (D)	1	4	0.04	0.42	0.01	0.03	0.01	<0.01			0.19
Site (S)	4	12	0.08	0.10	0.19	0.03	0.31	0.19			0.11
Plot (P)	12	0	nt	nt	nt	nt	nt	nt			nt
Mixing (M)	1	4	0.13	0.52	0.02	<0.01	0.07	0.13			0.51
S × M	4	12	0.03	0.28	0.09	0.39	0.78	0.09			0.65
D × M	1	4	0.05	0.03	0.09	0.83	0.46	0.51			0.19
P × M	12	0	nt	nt	nt	nt	nt	nt			nt
Seedling (G)	2	8	<0.01	0.00	<0.01	0.05	<0.01	<0.01			<0.01
S × G	8	24	0.70	0.78	0.56	0.11	0.12	0.37			0.06
D × G	2	8	<0.01	0.01	0.11	0.36	0.21	0.13			0.97
P × G	24	0	nt	nt	nt	nt	nt	nt			nt
M × G	2	8	0.01	0.23	0.04	0.73	0.68	0.03			0.05
S × M × G	8	24	0.97	0.80	0.97	0.10	0.31	0.71			0.79
D × M × G	2	8	<0.01	0.33	<0.01	0.30	0.87	0.96			0.45
S × M × G	24	0	nt	nt	nt	nt	nt	nt			nt
Error	0										
Total	107										

Note: The seedling response variables are shown in Figs. 2, 3, 4, and 6. RCD, root-collar diameter; df, degrees of freedom; nt, not tested.

Fig. 3. Relative increase in shoot dry mass (a) and root dry mass (b) and change in shoot/root dry mass ratio (c) of control (C), inoculated (I), and nursery (N) black spruce seedlings at the end of the second growing season. Bars are means, and error bars are SEs ($n = 108$). See Table 3 for ANOVA p values.



significant difference between these seedling types in insect-defoliated sites ($S = 1.3$, $p < 0.45$).

Shoot/root dry mass ratio

The largest increases in shoot/root ratio occurred in mixed plots of insect-defoliated sites for all of the seedling types (Fig. 3c, Table 3). In insect-defoliated sites, control and inoculated seedlings had significantly greater increases in shoot/root ratio in mixed plots compared with that in nonmixed plots ($S = 10.5$, $p < 0.01$ and $S = 6.0$, $p < 0.01$, respectively). In burned sites, only the nursery seedlings had a significantly greater increase in shoot/root ratio in mixed plots than in nonmixed plots ($S = 3.3$, $p < 0.01$). Inoculated seedlings were not significantly different in shoot/root ratio from the control seedlings, except in the mixed plots of insect-defoliated sites where control seedlings had a significantly greater increase than the inoculated seedlings ($S = 4.3$, $p < 0.01$).

Foliar nutrient concentration

At the end of the second growing season, foliar N, P, and K concentrations were significantly greater (49.8%, 28.5%, and 31.9%, respectively) for seedlings in insect-defoliated sites than those in burned sites ($F = 10.4$, $p < 0.03$; $F = 17.4$, $p < 0.01$; $F = 41.8$, $p < 0.003$; respectively) (Fig. 4, Table 3). Foliar N concentration of seedlings in forest floor mixed plots was significantly higher (32.2%) than in nonmixed plots ($F = 77.9$, $p < 0.001$; Fig. 4a). However, foliar P concentration of inoculated seedlings was significantly greater (19.3%) than that of control seedlings ($S = 4.5$, $p < 0.01$; Fig. 4b). Potassium concentration of inoculated seedlings was not significantly different from the control seedlings ($S = 0.1$, $p < 0.99$; Fig. 4c). Potassium concentration of nursery seedlings was greater in plots with mixed forest floors than those in nonmixed plots ($S = 4.2$, $p < 0.01$). However, for control and inoculated seedlings, there was no significant difference in K concentration between mixed and nonmixed plots.

Regression analyses of foliar N concentration on relative increase in shoot dry mass showed significant positive relationships for control, inoculated, and nursery-grown seedlings ($p \leq 0.001$ for each; Figs. 5a–5c). However, coefficients of determination indicated that foliar N concentration explained relatively little of the variability in shoot dry mass increase for the three seedling groups ($R^2 = 0.38$, 0.27, and 0.50, respectively).

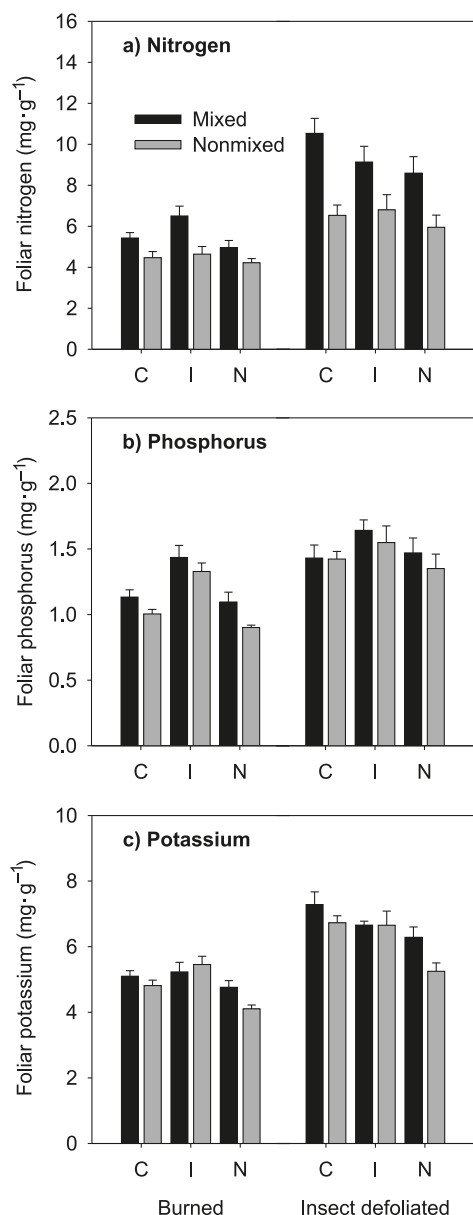
ECM formation

At the end of the first growing season, ECM formation (percentages of total short roots) was significantly greater (47%) in inoculated seedlings than in control seedling ($S = 16.6$, $p < 0.001$; Fig. 6a, Table 3). At the end of the second growing season, ECM formation was 27% greater in inoculated seedlings compared with the control ($S = 5.7$, $p < 0.001$; Fig. 6b, Table 3).

Soil physical properties

Soil moisture was significantly higher (48%) in mixed plots than in nonmixed plots ($F = 16.3$, $p < 0.016$; Table 4). Plots in burned sites had 64% deeper forest floor humus ($F = 6.4$, $p < 0.06$) with 33% higher soil moisture than

Fig. 4. Concentrations of foliar nitrogen (a), phosphorus (b), and potassium (c) of control (C), inoculated (I), and nursery (N) black spruce seedlings at the end of the second growing season. Bars are means, and error bars are SEs ($n = 108$). See Table 3 for ANOVA p values.

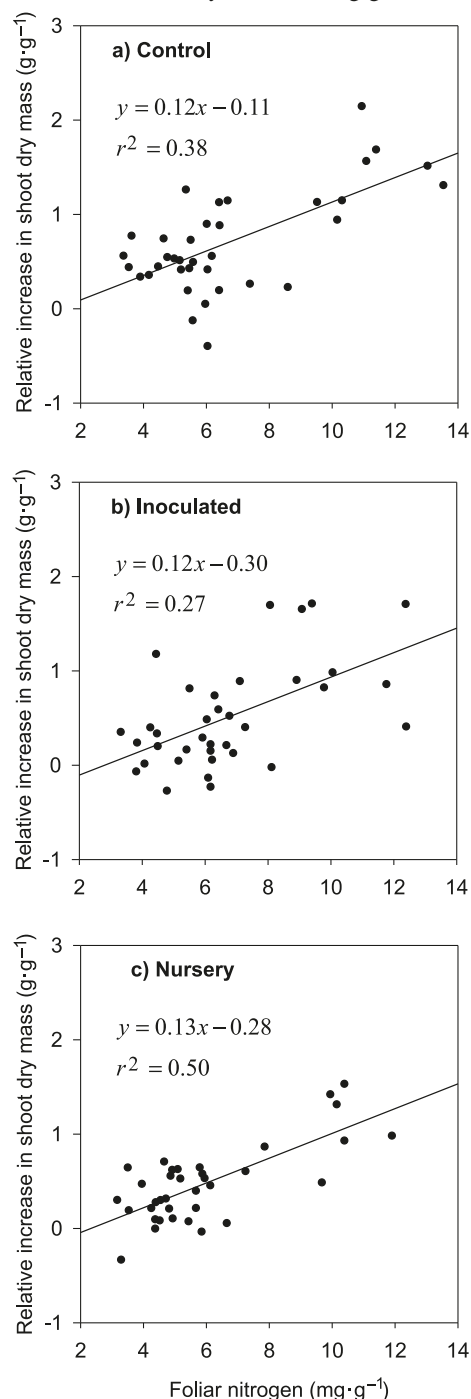


insect-defoliated sites ($F = 6.1$, $p < 0.07$; Table 4). Soil temperature did not vary significantly between the burned and insect-defoliated sites (Table 4).

Litter decomposition

After two growing seasons, litter mass loss was significantly higher (21.5%) in mixed plots than nonmixed plots ($F = 9.8$, $p < 0.035$; Fig. 7a). However, soil respiration was significantly higher (58%) in nonmixed plots than in mixed plots ($F = 29.7$, $p < 0.006$; Fig. 7b). Litter mass loss did not vary significantly between burned and insect-defoliated sites (Fig. 7a), but soil respiration was 51% higher in insect-defoliated sites than in burned sites ($F = 10.8$, $p < 0.03$; Fig. 7b).

Fig. 5. Relative increases in shoot dry mass at the end of the second growing season for control (a), inoculated (b), and nursery (c) black spruce seedlings as a function of foliar nitrogen concentration ($n = 36$). Negative values for relative growth are due to measurement error when there was very little seedling growth.



Soil chemical properties

After the second growing season, forest floors in non-mixed plots had significantly higher total N ($F = 15.8$, $p < 0.016$), C ($F = 105.9$, $p < 0.001$), K ($F = 7.1$, $p < 0.057$), and Ca ($F = 8.8$, $p < 0.041$) concentrations, as well as a higher C/N ratio ($F = 12.2$, $p < 0.025$) than the soil mixed plots (Table 5). Soil-mixed plots had significantly higher Fe

Fig. 6. Ectomycorrhiza (ECM) formation in control (C), inoculated (I), and nursery (N) black spruce seedlings at the end of the first (a) and second (b) growing seasons. Bars are means, and error bars are SEs (*n* = 108). See Table 3 for ANOVA *p* values.

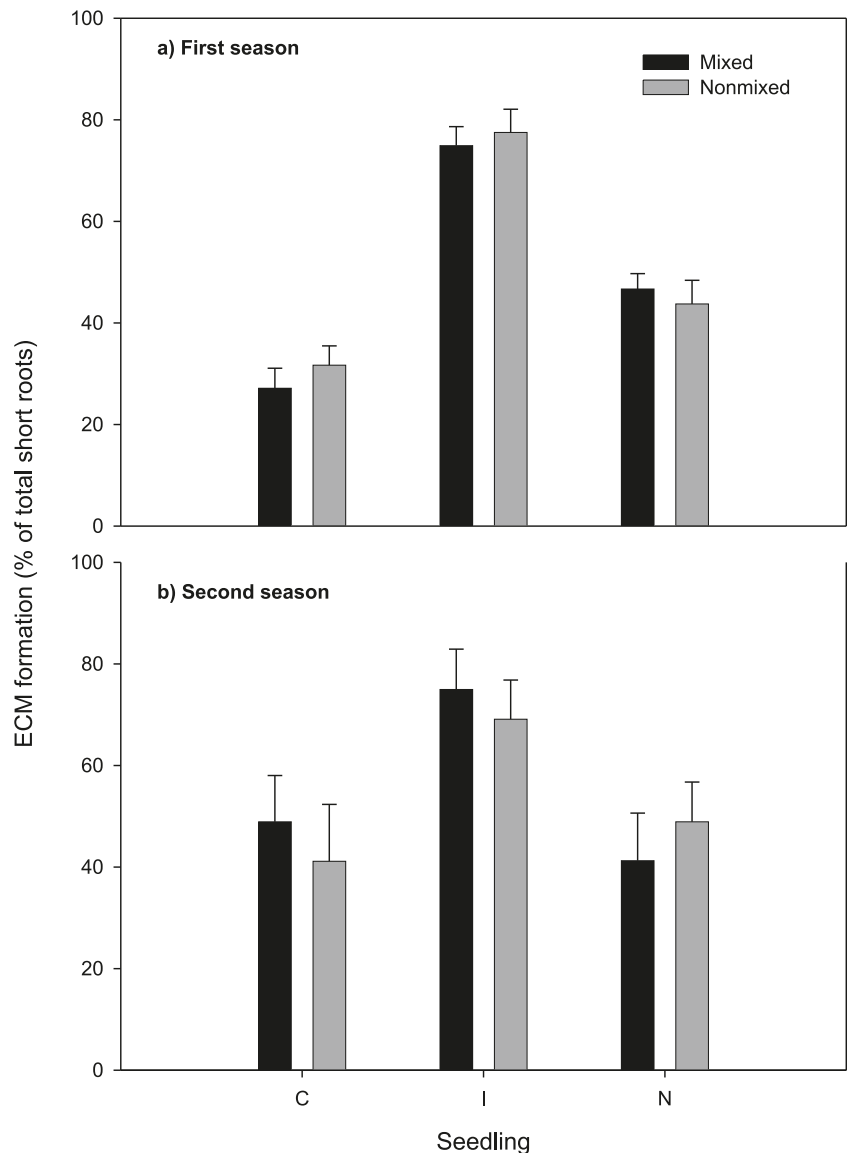


Table 4. Mean depth, temperature, and moisture of forest floor (humus) at the end of the second growing season in plots with mixed and nonmixed forest floors in burned and insect-defoliated sites (*n* = 36).

Disturbance and mixing	Depth (cm)	Temperature (°C)	Moisture (%)
Burned			
Mixed	12.8a	9.8a	46.5a
Nonmixed	15.6a	9.9a	34.0b
Insect defoliated			
Mixed	7.9b	12.2a	36.7a
Nonmixed	9.4b	11.3a	22.4b

Note: Within columns, values with different letters are significantly different at *p* < 0.05.

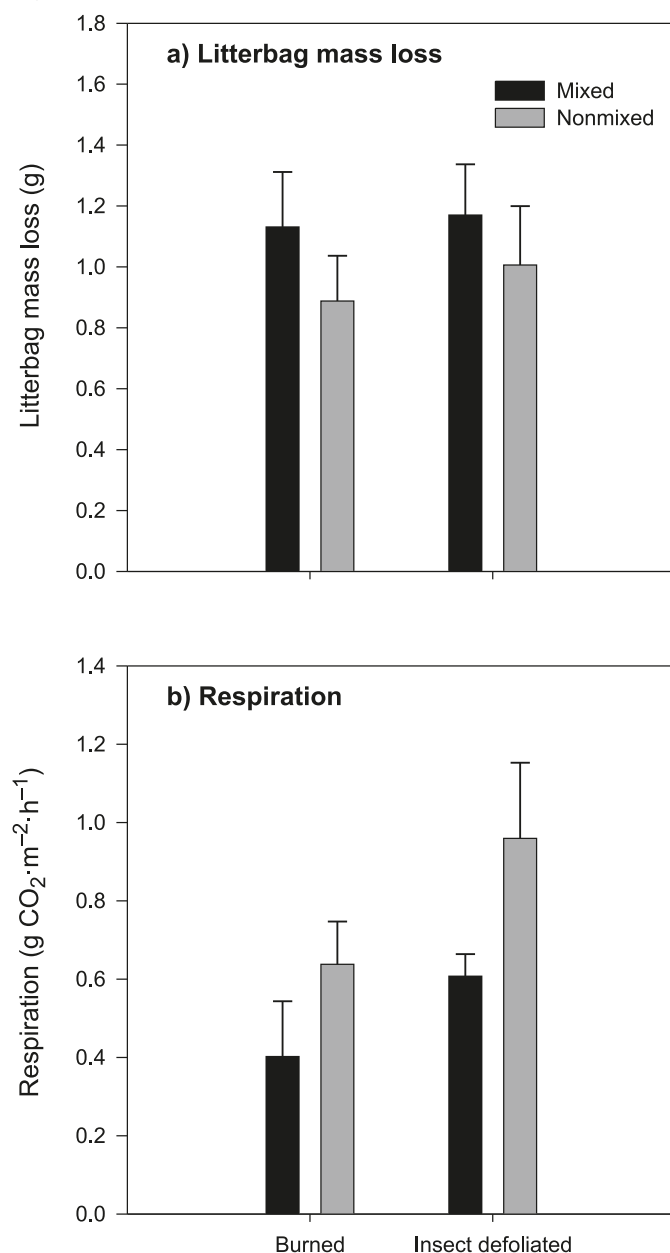
(*F* = 37.6, *p* < 0.004) and Al (*F* = 9.9, *p* < 0.035) concentrations, as well as higher pH (*F* = 27.6, *p* < 0.006), than nonmixed plots. The Ae horizon in the mixed plots had

significantly higher Al (*F* = 28.4, *p* < 0.006) concentrations than those in nonmixed plots (Table 5).

Forest floors in plots of burned sites had significantly higher total C (*F* = 14.9, *p* < 0.018), C/N ratio (*F* = 92.5, *p* < 0.001), Mg (*F* = 49.6, *p* < 0.002), and Na (*F* = 11.7, *p* < 0.027) than plots of insect-defoliated sites (Table 5). However, plots in insect-defoliated sites had significantly higher total N (*F* = 8.7, *p* < 0.042), Fe (*F* = 8.3, *p* < 0.045), total phenols (*F* = 8.2, *p* < 0.046), and Cu (*F* = 23.1, *p* < 0.009) than the plots in burned sites. Soils of the Ae horizon in plots of insect-defoliated sites had significantly higher total N (*F* = 6.9, *p* < 0.058), Al (*F* = 9.2, *p* < 0.039), and K (*F* = 9.5, *p* < 0.037) than those of the burned sites (Table 5). Soils in the B horizon of insect-defoliated sites had significantly higher pH (*F* = 25.9, *p* < 0.007), total N (*F* = 8.7, *p* < 0.042), K (*F* = 76.1, *p* < 0.001), and Cu (*F* = 8.0, *p* < 0.047) than those of the burned sites (Table 5).

Significant interaction effects between disturbance type and forest floor mixing occurred only with chemical proper-

Fig. 7. Litter mass loss (a) and soil respiration (b) at the end of the second growing season. Bars are means, and error bars are SEs ($n = 36$).



ties of the forest floors. At the end of the second growing season, the two-way interaction between disturbance type and forest floor mixing was significant for Fe ($F = 20.4$, $p < 0.011$), K ($F = 10.2$, $p < 0.033$), total C ($F = 30.0$, $p < 0.005$), and total N ($F = 19.2$, $p < 0.012$). Nonmixed plots of the insect-defoliated sites had significantly higher total N and K than the mixed plots (Table 5), and mixed plots had significantly higher Fe than nonmixed plots. In burned sites, there was no significant difference in total N, K, or Fe between the mixed and nonmixed plots. Total C was significantly higher in nonmixed plots than in mixed plots of both insect-defoliated and burned sites, but the difference was

Table 5. Soil chemical properties of forest floor (O), Ae, and B horizons at the end of the second growing season in plots with mixed and nonmixed forest floors in burned and insect defoliated sites.

Parameter and horizon	Insect defoliated		Burned	
	Mixed	Nonmixed	Mixed	Nonmixed
Ca ($\mu\text{g}\cdot\text{g}^{-1}$)				
O	1943.5ab	2742.0a	1224.6b	1737.2b
Ae	93.5a	70.8a	13.2b	12.8b
B	32.3a	17.3a	13.9a	15.6a
K ($\mu\text{g}\cdot\text{g}^{-1}$)				
O	409.3b	545.8a	523.5a	511.1a
Ae	39.9a	35.6a	16.4b	14.9b
B	51.0a	41.4ab	25.4b	22.9b
Mg ($\mu\text{g}\cdot\text{g}^{-1}$)				
O	521.9b	737.7b	1071.7a	1137.7a
Ae	25.8a	23.5a	9.7b	9.9b
B	16.0a	11.5	11.4a	10.7a
Na ($\mu\text{g}\cdot\text{g}^{-1}$)				
O	201.1b	67.1b	66.7a	46.4a
Ae	11.2a	11.7a	7.9a	8.9a
B	15.4a	12.8a	16.5a	14.4b
Al ($\text{mg}\cdot\text{g}^{-1}$)				
O	1461.1a	470.5b	703.7ab	486.5b
Ae	0.7a	0.5b	0.4c	0.3c
B	7.5ab	9.7a	5.6b	5.1b
Cu ($\mu\text{g}\cdot\text{g}^{-1}$)				
O	1.4a	1.2ab	0.8b	0.8b
Ae	0.76a	0.70a	0.43a	0.76a
B	1.53a	1.44a	0.72b	0.73b
Fe ($\mu\text{g}\cdot\text{g}^{-1}$)				
O	201.1a	67.1b	66.7b	46.4b
Ae	194.9a	83.4a	135.8a	73.9a
B	502.6a	590.4a	944.2a	750.4a
Zn ($\mu\text{g}\cdot\text{g}^{-1}$)				
O	29.1a	33.6a	15.5b	21.6ab
Ae	1.3ab	1.7a	0.6c	0.8bc
B	1.5a	1.6a	1.2a	1.1a
Phenols ($\text{mg}\cdot\text{g}^{-1}$)				
O	2.9ab	3.5a	1.6b	2.1b
Ae	0.81a	1.09a	0.86a	0.90a
B	3.33a	3.86a	2.87a	3.39a
pH				
O	4.3a	4.0b	4.2b	4.0b
Ae	4.18b	4.13b	4.33a	4.38a
B	4.83a	4.66b	4.50bc	4.46c
PO ₄ ($\mu\text{g}\cdot\text{g}^{-1}$)				
O	44.5a	54.1a	39.6a	52.1a
Ae	8.4a	8.4a	8.8a	6.7a
B	17.1a	17.0a	14.6a	17.7a
NH ₄ ($\mu\text{g}\cdot\text{g}^{-1}$)				
O	53.6a	53.7a	53.8a	45.2a
Ae	11.5a	12.9a	11.7a	11.9a
B	16.9a	16.0a	15.6a	17.9a
Total N ($\text{mg}\cdot\text{g}^{-1}$)				
O	9.6b	13.1a	9.3b	9.1b
Ae	0.61a	0.61a	0.28b	0.25b
B	1.63ab	1.92a	1.02b	1.02b

Note: Values are means ($n = 36$). Within rows, values with different letters are significantly different at $p < 0.05$.

much greater in insect-defoliated sites than in the burned sites (Table 5).

Discussion

Seedling response

Forest floor mixing resulted in enhanced black spruce seedling growth in the insect-defoliated sites but not in the burned sites. Height growth of control seedlings in mixed plots of insect-defoliated sites (mean 8.2 cm) exceeded the estimated optimal growth for 2-year-old black spruce (7 cm) reported by Malik and Timmer (1996). This rapid growth rate suggests that, in insect-defoliated sites, forest floor mixing may fully release seedlings from the below-ground effects of *Kalmia* dominance and significantly hasten canopy closure. In both insect-defoliated and burned sites, foliar N concentration was higher for seedlings in the forest floor mixed plots than nonmixed plots over the two growing seasons. This is an indication that soil mixing resulted in increased nutrient availability, likely because of improved soil pH, increased mineralization, and decreased belowground competition and nutrient interference.

Superior growth and foliar nutrient concentrations of seedlings in the insect-defoliated sites relative to burned sites was likely due to better litter quality (lower C/N ratio) resulting from partial canopy cover from the remnant trees. Bloom and Mallik (2004) found that litter quality is better in *Kalmia*-dominated sites with partial canopy cover than that with open canopy. They found that forest cover increases soil respiration, which in turn results in higher litter decomposition and soil nutrient availability. One would expect that soils in the insect-defoliated sites would be lower in concentrations of total phenols than those in the burned sites because total phenol concentration in plants increases with increased ultraviolet light exposure (Hattenschwiler and Vitousek 2000). However, large amounts of *Kalmia* phenols in the burned sites may have been adsorbed in charcoal (DeLuca and Aplet 2008) and on organic matter (Rice 1984) and, thus, were undetectable. Organic matter total C was significantly higher in burned sites than in insect-defoliated sites.

Regression analyses of seedling growth as a function of foliar N concentration over all sites in this study suggest that a substantial portion of the variability in seedling growth was not explained by nutrient availability. We speculate that canopy cover may also have important effects on aboveground factors related to seedling growth, which may explain why seedling growth was greater in insect-defoliated sites but limited in burned sites. Jobidon (1994) reported that maximum shoot growth for black spruce seedlings occurs at 60% full light. Photodamage to plant tissues can occur in full light, but spruce seedlings under partial canopy cover are not affected (Ronco 1975).

At the time of outplanting, the inoculated seedlings had greater ECM formation than the control seedlings. Two years later at the end of the experiment, the inoculated seedlings had higher level of ECM short roots, but they failed to have greater relative increases in shoot and root growth and dry biomass than the control seedlings. Finding no growth enhancement of the inoculated seedlings, we did not morphotype their ECM fungi to assess the persistence of *P. involutus*. It is not clear why the inoculated seedlings in this study did not achieve superior growth. It is unlikely that the

small size of the inoculated seedlings relative to the controls was important because seedling height at outplanting is not a conclusive indicator of growth. Stenstrom and Ek (1990) found that inoculated seedlings that were initially considerably smaller than control seedlings could have faster growth rates. Paterson (1997) found that larger seedlings had greater growth than smaller seedlings but only when the larger seedlings were comparable in height with competing vegetation, which was not the case in our study. The reduced growth of inoculated seedlings was not likely a function of nutrient uptake differences because the inoculated seedlings had higher foliar P and similar N and K compared with the control seedlings. Given that the inoculated seedlings had substantially higher ECM formation than the controls seedlings after outplanting, it is possible that early growth of inoculated seedlings was limited because most energy might have been used for mycelial growth. It is common for ECM to benefit seedlings only during specific stages of seedling development (Perry et al. 1987). Browning and Whitney (1992) found that black spruce was slow to respond to inoculation compared with jack pine. Because we did not morphotype the ECM of the outplanted seedlings at the end of the experiment, we cannot confirm the persistence of *P. involutus* and attribute its beneficial effects on P uptake. A more extensive field study on the ability of *P. involutus* to benefit black spruce seedlings in *Kalmia*-dominated soils should be undertaken, in which the formation and persistence of inoculated and noninoculated ECM fungal species or strains, rhizosphere soil chemistry, and seedling growth are followed over 5–10 years.

Habitat response to soil mixing

Decomposition rates, as indicated by litter mass loss, were higher in soil-mixed plots than in nonmixed plots. Burger and Pritchett (1988) found increased decomposition after vegetation clearing and harrowing. Concentrations of available N and P (NH₄-N and PO₄-P) were not higher in mixed forest floors than in nonmixed plots. This result was anticipated because increased mineralization is often accompanied by increased plant uptake and immobilization in boreal soils (Vitousek and Matson 1985). Frey et al. (2003) also found that soil mixing did not result in an increase in available N and suggested that this occurred because mixing introduced litter containing labile C, which in turn caused immobilization by microbes. Ugolini et al. (1990) found that forest floor mixing improved decomposition in coastal Alaskan forest.

Soil respiration and litter mass loss (hence, decomposition rate) were higher in insect-defoliated plots than in plots of burned sites with concomitant increase the seedling growth and foliar nutrient concentrations. The enhanced decomposition rates in insect-defoliated plots also correspond to higher concentrations of total phenols and tannins, which may have been a C source for increased microbial activity (Bradley et al. 1997; Kraus 2002). However, concentrations of available N and P (NH₄-N and PO₄-P) were not higher in plots of insect-defoliated than the burned sites. This was likely because increased mineralization of these nutrients may have been balanced by plant uptake and immobilization in microbial biomass or by leaching and volatilization, which is commonly associated with increases in microbial activity (Vitousek and Matson 1985). Higher indices of available

C (basal respiration and microbial biomass) in *Kalmia*-dominated than *Kalmia*-eradicated plots reported by LeBel et al. (2008) support this explanation.

High *Kalmia* phenol in mineral soils was reported to decrease the concentration of total N and increase the concentrations of Fe, Mn, Al, and PO₄-P (Inderjit and Mallik 1996). Black spruce seedlings grown in *Kalmia* soils had high foliar concentrations of Ca, Al, Fe, and K (Mallik 2001) and a lower concentration of N (Yamasaki et al. 1998). In our study, higher levels of total phenols in insect-defoliated sites were not accompanied by lower concentration of total N, but the concentration of Fe was higher. Mixed forest floors had higher pH compared with nonmixed forest floors. Similarly, Keenan et al. (1994) found decreased acidity after forest floor mixing in the coastal forests of British Columbia, which they attributed to mixing mineral soil of low acidity into organic matter of high acidity. In our study, some mineral soil was mixed into the forest floor, which may have been enough to raise the pH. Decreased acidity in soils generally results in enhanced nutrient availability.

Conclusions and management implications

Kalmia-dominated soils in sites originating from insect defoliation have better physical and chemical conditions for black spruce growth than those originating from nonsevere wildfires. Mixing the forest floor of *Kalmia*-dominated sites results in improved soil physical and chemical properties relative to that of nonmixed forest floor. Therefore, forest floor mixing as a reforestation approach would likely enhance reestablishment of black spruce on insect-defoliated sites that have partial canopy cover but not on burned sites that lack canopy cover. Inoculating black spruce seedlings with *P. involutus* would likely result in high ECM formation but may not improve the early growth of seedlings on *Kalmia*-dominated sites. However, this needs further investigation.

Acknowledgements

We thank Robert Bradley, Qing-Lai Dang, and three anonymous reviewers for their helpful comments on an earlier version of this paper. This study received partial financial and logistical support from Parks Canada and a Natural Sciences and Engineering Research Council discovery grant awarded to the second author. We thank the staff of Terra Nova National Park for logistical help during the fieldwork. Leonard Hutchinson and Ken Brown helped with the experimental design, seedling inoculation, and statistical analyses.

References

- Allen, S.E. 1989. Chemical analysis of ecological materials. Blackwell Scientific Publications, London.
- Bledsoe, C.S., Tennyson, K., and Lopushinsky, W. 1982. Survival and growth of outplanted Douglas-fir seedlings inoculated with mycorrhizal fungi. *Can. J. For. Res.* **12**: 720–723. doi:10.1139/x82-109.
- Bloom, R., and Mallik, A.U. 2004. Indirect effects of black spruce (*Picea mariana*) cover on community structure and function in sheep laurel (*Kalmia angustifolia*) dominated heath in eastern Canada. *Plant Soil*, **265**: 279–293. doi:10.1007/s11104-005-0508-4.
- Bradley, R.L., Titus, B.D., and Fyles, J.W. 1997. Nitrogen acquisition and competitive ability of *Kalmia angustifolia* L., paper birch (*Betula papyrifera* Marsh.) and black spruce (*Picea mariana* (Mill.) B.S.P.) seedlings grown on different humus forms. *Plant Soil*, **195**: 209–220. doi:10.1023/A:1004263716346.
- Browning, M.H.R., and Whitney, R.D. 1992. Field performance of black spruce and jack pine inoculated with selected species of ectomycorrhizal fungi. *Can. J. For. Res.* **22**: 1974–1982. doi:10.1139/x92-168.
- Burger, J.A., and Pritchett, W.L. 1988. Site preparation effects on soil moisture and available nutrients in a pine plantation in the Florida flatwoods. *For. Sci.* **34**: 77–87.
- Danielson, R.M. 1982. Taxonomic affinities and criteria for identification of the common ectomycorrhizal symbiont of pines. *Can. J. Bot.* **60**: 7–18. doi:10.1139/b82-002.
- Danielson, R.M., Griffiths, C.L., and Parkinson, D. 1984. Effects of fertilization on the growth and mycorrhizal development of container-grown jack pine seedlings. *For. Sci.* **30**: 828–835.
- DeLuca, T.H., and Aplet, G.H. 2008. Charcoal and carbon storage in forest soils of the Rocky Mountain West. *Front. Ecol. Environ.* **6**(1): 18–24. doi:10.1890/070070.
- Frey, B.R., Lieffers, V.J., Munson, A.D., and Blenis, P.V. 2003. The influence of partial harvesting and forest floor disturbance on nutrient availability and understory vegetation in boreal mixedwoods. *Can. J. For. Res.* **33**: 1180–1188. doi:10.1139/x03-042.
- Hattenschwiler, S., and Vitousek, P.M. 2000. The role of polyphenols in terrestrial nutrient cycling. *Trends Ecol. Evol.* **15**: 238–243. doi:10.1016/S0169-5347(00)01861-9. PMID:10802549.
- Hoffmann, W.A., and Poorter, H. 2002. Avoiding bias in calculations of relative growth rate. *Ann. Bot. (Lond.)*, **90**: 37–42. doi:10.1093/aob/mcf140.
- Inderjit, I., and Mallik, A.U. 1996. The nature of interference potential of *Kalmia angustifolia*. *Can. J. For. Res.* **26**: 1899–1904. doi:10.1139/x26-214.
- Ingleby, K., Mason, P.A., Last, F.T., and Fleming, L.V. 1990. Identification of ectomycorrhizas. Her Majesty's Stationery Office, London. ITE Res. Publ. 5.
- Joanisse, G.D., Bradley, R.L., Preston, C.M., and Munson, A.D. 2007. Soil enzyme inhibition by condensed litter tannins may drive ecosystem structure and processes: the case of *Kalmia angustifolia*. *New Phytol.* **175**: 535–546. doi:10.1111/j.1469-8137.2007.02113.x. PMID:17635228.
- Jobidon, R. 1994. Light threshold for optimal black spruce (*Picea mariana*) growth and development under brush competition. *Can. J. For. Res.* **24**: 1629–1635. doi:10.1139/x94-211.
- Kalra, Y.P., and Maynard, D.G. 1991. Methods manual for forest soil and plant analysis. For. Can. North. For. Cent. Inf. Rep. NOR-S-319.
- Keenan, R.J., Messier, C., and Kimmins, J.P. 1994. Effects of clearcutting and soil mixing on soil properties and understory biomass in western red cedar and western hemlock forests on northern Vancouver Island, Canada. *For. Ecol. Manage.* **68**: 251–261. doi:10.1016/0378-1127(94)90049-3.
- Kraus, T.E.C. 2002. Tannins and nutrient dynamics in forest soils: plant–litter–soil interactions. Ph.D. thesis, University of California, Davis, Calif.
- Landis, T.D. 1990. Atmospheric environment. Vol. 3. In *The container tree nursery manual*. Edited by T.D. Landis, Titus, R.W., McDonald, S.E., and Barnett, J.P. Agricultural Handbook 674. USDA Forest Service, Washington, D.C. pp. 7–45.
- LeBel, P., Thiffault, N., and Bradley, R.L. 2008. *Kalmia* removal increases nutrient supply and growth of black spruce seedlings: an effect fertilizer cannot emulate. *For. Ecol. Manage.* **256**: 1780–1784. doi:10.1016/j.foreco.2008.02.050.
- Malik, V., and Timmer, V.R. 1996. Growth, nutrient dynamics, and interspecific competition of nutrient-loaded black spruce seed-

- lings on a boreal mixedwood site. *Can. J. For. Res.* **26**: 1651–1659. doi:10.1139/x26-186.
- Mallik, A.U. 1991. Cutting, burning, and mulching to control *Kalmia*: results of a greenhouse experiment. *Can. J. For. Res.* **21**: 417–420. doi:10.1139/x91-054.
- Mallik, A.U. 1995. Conversion of temperate forests into heaths: role of ecosystem disturbance and ericaceous plants. *Environ. Manage.* **19**: 675–684. doi:10.1007/BF02471950.
- Mallik, A.U. 2001. Black spruce growth and understory species diversity with and without sheep laurel. *Agron. J.* **93**: 92–98.
- Mallik, A.U., Zhu, H., and Park, Y.G. 1998. Overcoming *Kalmia*-induced growth inhibition of *Picea mariana* by mycorrhizal inoculation. *J. Korean For. Soc.* **87**: 429–444.
- Marx, D.H., and Cordell, C.E. 1988. Specific ectomycorrhiza improve reforestation and reclamation in eastern United States. In *Proceedings of the Canadian Workshop on Mycorrhizae in Forestry*, 1–4 May 1988, Sainte-Foy, Que. Edited by M. Lalonde, and Y. Piché. Centre de recherche en biologie forestière, Faculté de foresterie et de géodésie, Université Laval, Sainte-Foy, Que.
- Meades, W.J. 1986. Successional status of ericaceous dwarf shrub heath in eastern Newfoundland. Ph.D. thesis, University of Connecticut, Storrs, Conn.
- Meades, W.J., and Moores, L. 1989. Forest site classification manual: a field guide to the Damman forest types of Newfoundland. Western Newfoundland Model Forest, Corner Brook, N.L. For. Resour. Dev. Agree. Rep. 003.
- Messier, C., Keenan, R.J., and Kimmins, J.P. 1995. The effects of soil mixing on soil nutrient status, recovery of competing vegetation and conifer growth on cedar–hemlock cutovers in coastal British Columbia. *New For.* **9**: 163–179.
- Molina, R., and Chamard, J. 1983. Use of the ectomycorrhizal fungus *Laccaria laccata* in forestry. II. Effects of fertilizer forms and levels on ectomycorrhizal development and growth of container-grown Douglas-fir and ponderosa pine seedlings. *Can. J. For. Res.* **13**: 89–95. doi:10.1139/x83-013.
- Newton, P.F., and Weetman, G.F. 1993. Stand density management diagram and their development and utility in black spruce management. *For. Chron.* **69**: 421–430.
- Paterson, J. 1997. Growing environment and container type influences field performance of black spruce container stock. *New For.* **13**: 329–339.
- Perry, D.A., and Choquette, C.C. 1987. Allelopathic effects on mycorrhizae. In *Allelopathy, its role in agriculture, forestry, and ecology*. Edited by G. Waller. American Chemical Society, Washington, D.C. Am. Chem. Soc. Symp. Ser. 330. pp. 185–194.
- Perry, D.A., Molina, R., and Amaranthus, M.P. 1987. Mycorrhizae, mycorrhizospheres, and reforestation: current knowledge and research needs. *Can. J. For. Res.* **17**: 929–940. doi:10.1139/x87-145.
- Power, R.G. 1996. Forest fire history and vegetation analysis of Terra Nova National Park. Technical report. Parks Canada Agency, Terra Nova National Park, Glovertown, N.L.
- Prescott, C.E., Maynard, D.G., and Laiho, R. 2000. Humus in northern forests: friend or foe? *For. Ecol. Manage.* **133**: 23–36. doi:10.1016/S0378-1127(99)00295-9.
- Prévost, M. 1994. Scalping and burning of *Kalmia angustifolia* (Ericaceae) litter: effects on *Picea mariana* establishment and ion leaching in a greenhouse experiment. *For. Ecol. Manage.* **63**: 199–218. doi:10.1016/0378-1127(94)90111-2.
- Rice, E.L. 1984. Allelopathy. 2nd ed. Academic Press, Inc., New York.
- Ronco, F. 1975. Diagnosis: “sunburned” trees. *J. For.* **73**: 31–35.
- Rowe, J.S. 1972. Forest regions of Canada. Canadian Forestry Service, Department of Environment, Ottawa, Ont.
- Sakakibara, S.M., Jones, M.D., Gillespie, M., Hagerman, S.M., Forrest, M.E., Simard, S.W., and Durall, D.M. 2002. A comparison of ectomycorrhiza identification based on morphotyping and PCR–RFLP analysis. *Mycol. Res.* **106**: 868–878. doi:10.1017/S0953756202006263.
- Soil Classification Working Group. 1998. The Canadian system of soil classification. 3rd ed. Agriculture and Agri-Food Canada, Ottawa, Ont. Publ. 1646.
- Stenström, E., and Ek, M. 1990. Field growth of *Pinus sylvestris* following nursery inoculation with mycorrhizal fungi. *Can. J. For. Res.* **20**: 914–918. doi:10.1139/x90-123.
- Swain, T., and Hillis, W.E. 1959. The phenolic constituents of *Prunus domestica* I — the quantitative analysis of phenolic constituents. *J. Sci. Food Agric.* **10**: 63–68. doi:10.1002/jsfa.2740100110.
- Thiffault, N., and Jobidon, R. 2006. How to shift unproductive *Kalmia angustifolia*–*Rhododendron groenlandicum* heath to productive conifer plantation. *Can. J. For. Res.* **36**: 2364–2376. doi:10.1139/X06-090.
- Thiffault, N., Titus, B.D., and Munson, A.D. 2004. Black spruce seedlings in a *Kalmia*–*Vaccinium* association: microsite manipulation to explore interactions in the field. *Can. J. For. Res.* **34**: 1657–1668. doi:10.1139/x04-046.
- Thiffault, N., Titus, B.D., and Munson, A.D. 2005. Silvicultural options to promote seedling establishment on *Kalmia*–*Vaccinium*-dominated sites. *Scand. J. For. Res.* **20**: 110–121. doi:10.1080/02827580510008356.
- Ugolini, F.C., Bormann, B.T., and Bowers, F.H. 1990. The effect of tree throw on forest soil development in southeast Alaska. USDA Forest Service, Pacific Northwest Research Station, Portland, Ore.
- Villeneuve, N., Le Tacon, F., and Bouchard, D. 1991. Survival of inoculated *Laccaria bicolor* in competition with native ectomycorrhizal fungi and effects on the growth of outplanted Douglas-fir seedlings. *Plant Soil*, **135**: 95–107. doi:10.1007/BF00014782.
- Vitousek, P.M., and Matson, P.A. 1985. Disturbance, nitrogen availability, and nitrogen losses in an intensively managed loblolly pine plantation. *Ecology*, **66**: 1360–1376. doi:10.2307/1939189.
- Walker, G.R. 2005. Black spruce regeneration in *Kalmia* dominated sites: effects of mycorrhizal inoculation and forest floor mixing. M.Sc. thesis, Department of Biology, Lakehead University, Thunder Bay, Ont.
- Wallstedt, A., Coughlan, A., Munson, A.D., Nilsson, M.-C., and Margolis, H.A. 2002. Mechanisms of interaction between *Kalmia angustifolia* cover and *Picea mariana* seedlings. *Can. J. For. Res.* **32**: 2022–2031. doi:10.1139/x02-124.
- Yamasaki, S.H., Fyles, J.W., Egger, K.N., and Titus, B.D. 1998. The effect of *Kalmia angustifolia* on the growth, nutrition, and ectomycorrhizal symbiont community of black spruce. *For. Ecol. Manage.* **105**: 197–207. doi:10.1016/S0378-1127(97)00285-5.
- Yamasaki, S.H., Fyles, J.W., and Titus, B.D. 2002. Interactions among *Kalmia angustifolia*, soil characteristics, and the growth and nutrition of black spruce seedlings in two boreal Newfoundland plantations of contrasting fertility. *Can. J. For. Res.* **32**: 2215–2224. doi:10.1139/x02-119.
- Zeng, R.-S., and Mallik, A.U. 2006. Selected ectomycorrhizal fungi of black spruce (*Picea mariana*) can detoxify phenolic compounds of *Kalmia angustifolia*. *J. Chem. Ecol.* **32**: 1473–1489. doi:10.1007/s10886-006-9063-6. PMID:16718563.
- Zeng, R.S., Mallik, A.U., and Setliff, E. 2003. Growth stimulation of ectomycorrhizal fungi by root exudates of Brassicaceae plants: role of degraded compounds of indole glucosinolates. *J. Chem. Ecol.* **29**: 1337–1355. doi:10.1023/A:1024257218558. PMID:12918920.